

QICN v22 Internal Note: Independent Finite Separator-Complete Incidence Package and Conditional Closure of I_{int}

QICN internal formal-audit scaffold

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Governance boundary

This document does not certify external support, consciousness, phenomenality, identity transfer, or bridge-burden closure. It is an internal mathematical and implementation scaffold. It does not report human-subject data, external adjudication, reviewer certification, or empirical validation. The package constructed in this note is finite and synthetic; it may be used as a conditional formal witness only after its catalog-completeness premise is independently accepted.

1 Scope and layer separation

Ontology. A system S is treated only as an admissible object with histories, identity data, intervention-response coordinates, and separator tests. No consciousness, agency, subjecthood, or phenomenal identity is attributed.

Mathematical model. The relevant structure is a finite bipartite incidence graph $G_S = (V_\Theta \sqcup V_R, E)$ plus an explicit finite factorization-cut universe.

Implementation. The finite package is represented by `docs/theory/FINITE_SEPARATOR_COMPLETE_PACKAGE_v22.json`; it is audited by `scripts/audit-finite-separator-package.js`. The product-separator negative control is represented separately.

Language. “Separator-complete”, “response-operative”, and “connected” are formal predicates over the finite package. They are not synonyms for consciousness or identity transfer.

Interpretation. The result is aggressive but bounded: accepted finite separator-complete connected incidence implies atomicity relative to the declared finite factorization universe, and therefore conditional closure of I_{int} . It is not a global proof from rigid identity plus continuity plus fidelity alone.

2 Corpus anchors

This note extends the v20 closure note and is anchored to the monolithic Paper 5 claims. Proposition 5.37 is the original integration-transfer anchor, and Theorem 5.11 is the prior conditional theorem under an atomic separator. The formal style follows the framework’s Paper 5 surface [Per26], categorical factorization language [Mac98], and finite graph reasoning compatible with causal graph

vocabulary [Pea09]. These references orient the formalism only; they do not externally validate QICN.

3 Finite package primitives

Definition 3.1 (Finite response-separator incidence package). *A finite response-separator incidence package for a system presentation S is a tuple*

$$\text{Cert}(S) = (V_\Theta, V_R, E, \mathcal{P}, \mathcal{E}, \mathcal{C}),$$

where:

- (i) V_Θ is a finite set of elementary separator vertices declared by a separator catalog $\mathcal{C}$;
- (ii) V_R is a finite set of elementary intervention-response coordinate vertices;
- (iii) $E \subseteq V_\Theta \times V_R$ is the incidence relation;
- (iv) $\mathcal{P}$ is a finite set of typed perturbation records witnessing the edges of E ;
- (v) $\mathcal{E}$ is an enumeration policy for non-trivial binary factorization cuts over $V_\Theta \sqcup V_R$.

An edge $(\theta, r) \in E$ is admissible only if its perturbation record preserves the declared identity typing and history typing while producing a positive separator-margin change.

Definition 3.2 (Independent construction boundary). *The package is independently constructed when V_Θ , V_R , E , and $\mathcal{P}$ are declared before applying the atomicity predicate, and the construction rule does not query whether Θ_S is atomic. Operationally, the v22 package is generated from a literal separator catalog, response-coordinate catalog, and perturbation table. The audit then checks graph connectedness and cut obstruction afterward.*

Definition 3.3 (Finite separator-completeness). *A package $\text{Cert}(S)$ is finite separator-complete relative to its declared universe when:*

- (i) every declared separator in the catalog $\mathcal{C}$ appears exactly as a vertex of V_Θ ;
- (ii) every $\theta \in V_\Theta$ is response-operative, meaning it has at least one incident edge witnessed by a positive typed perturbation record;
- (iii) every $r \in V_R$ has at least one incident edge;
- (iv) every admissible factorization considered by the finite model induces a binary cut listed by the enumeration policy $\mathcal{E}$;
- (v) a factor-local decomposition corresponds to a non-trivial cut with no incidence edge crossing the cut.

This is a finite relative condition. It does not claim completeness over undeclared separators, unrecorded response coordinates, infinite systems, or external empirical systems.

Definition 3.4 (Connected finite incidence). *The incidence graph of $\text{Cert}(S)$ is connected when the undirected bipartite graph $G_S = (V_\Theta \sqcup V_R, E)$ has exactly one connected component after rejecting isolated non-operative vertices.*

Definition 3.5 (Relative atomicity). *The separator family is atomic relative to $\text{Cert}(S)$ when no non-trivial finite factorization cut enumerated by $\mathcal{E}$ is factor-local, i.e. every enumerated non-trivial cut has at least one crossing incidence edge.*

Definition 3.6 (Hash-stable package integrity). *A finite package is hash-stable when the declared package digest equals*

$$\text{SHA256}(\text{stable_json}(P \setminus \{\text{package_sha256}\})).$$

The digest excludes only the top-level digest field itself. Declared audit metadata, admission rules, perturbation records, and limitations remain inside the digest. This avoids the self-referential hash defect in which recomputing a digest over a package that already contains its own digest changes the object being hashed.

4 The constructed v22 package

Criterion 4.1 (v22 connected finite package). *The file `FINITE_SEPARATOR_COMPLETE_PACKAGE_v22.js on` declares six separator vertices, five response-coordinate vertices, twelve typed incidence edges, and twelve typed perturbation records. The audit enumerates all canonical binary cuts over the eleven vertices, restricts to non-trivial factor cuts containing at least one separator and one response on each side, and verifies that no such cut is edge-closed. It also stores the recomputed counts inside the package as declared audit metadata and verifies that the package hash is stable under the top-level digest-exclusion rule.*

Criterion 4.2 (Product-separator negative control). *The file `FINITE_SEPARATOR_COMPLETE_PACKAGE_PRODUCT_NEGATIVE_CONTROL_v22.json` deliberately declares two disconnected product components. The same audit must reject it as an atomicity witness. This prevents the validator from merely accepting any syntactically complete finite package.*

5 Main theorem

Theorem 5.1 (Finite separator-complete connected incidence implies relative atomicity). *Let $\text{Cert}(S)$ be a finite separator-complete response-separator incidence package. If its incidence graph is connected, then Θ_S is atomic relative to the finite factorization universe declared by $\text{Cert}(S)$.*

Proof. Assume for contradiction that Θ_S is not atomic relative to $\text{Cert}(S)$. Then there exists an enumerated non-trivial factorization cut

$$V_\Theta \sqcup V_R = A \sqcup B$$

with both A and B containing at least one response-operative separator and at least one response-coordinate vertex, and with no edge of E crossing from A to B .

Because the cut is non-trivial and all represented separators and response vertices are operative, both induced subgraphs are non-empty. The absence of crossing edges means that the graph decomposes as the disjoint union of the subgraph induced by A and the subgraph induced by B . Hence G_S has at least two connected components. This contradicts connectedness. Therefore no enumerated non-trivial factorization cut is factor-local, which is exactly relative atomicity for the finite package. $\square$

Corollary 5.2 (v22 finite-witness conditional closure of I_{int}). *If the v22 finite package is accepted as separator-complete for the declared finite system model, if its audit passes, and if the factorization-loss terms are faithful to history, identity, response, and gauge preservation, then $I_{\text{int}}(S)$ exists and is unique as a gauge-invariant binary factorization-gap invariant relative to that finite model.*

Proof. The audit-passing finite package gives connected finite incidence. The previous theorem gives relative atomicity. The prior conditional theorem under atomic separator, anchored by Theorem 5.11, then applies to the finite model: a zero-loss non-trivial factorization would require a factor-local separator cut, but relative atomicity forbids such a cut. Therefore the factorization-gap predicate defining I_{int} is well-defined and invariant under admissible gauge transformations that preserve the finite package. Uniqueness follows from equality of the binary zero-set for zero-loss factorizations. $\square$

6 Non-circularity and reviewer burden

Remark 6.1 (Reviewer burden). *A reviewer must verify that the separator catalog and response-coordinate catalog can be obtained operationally without first proving atomicity. The admissible workflow is: declare elementary separator tests, declare elementary response coordinates, record typed perturbation evidence, construct the incidence graph, enumerate cuts, and only then apply connectedness. The inadmissible workflow is: assume no factor-local separator partition exists and then build the catalog around that assumption. If the inadmissible workflow is used, the v22 certificate is circular and must be rejected.*

Remark 6.2 (What this closes). *The v22 construction closes a finite, implementation-checkable route: accepted finite separator-complete connected incidence implies relative atomicity, which gives conditional I_{int} closure. It does not close the global atomic-separator lemma for arbitrary continuous or infinite systems, and it does not transform synthetic internal evidence into empirical validation.*

Remark 6.3 (Negative control interpretation). *The product-separator negative control fails the connectedness and factor-local cut audit. This is the expected behavior. It shows that the validator distinguishes a connected finite witness from a product-decomposed incidence graph. It does not show that real systems satisfy the positive package.*

7 Implementation status

The implementation consists of the package constructor, validator, positive package, negative control package, independent catalog protocol, blinding protocol, double-blind review packet, and generated audit report. The validator is deterministic and computes a package hash that excludes only the top-level digest field. The full result is recorded in `docs/reports/FINITE_SEPARATOR_COMPLETE_PACKAGE_v22_AUDIT.json`. Promotion to a source paper or theorem registry entry still requires human mathematical curation and an explicit decision about the finite-universe premise.

References

- [Mac98] Saunders Mac Lane. *Categories for the Working Mathematician*. 2nd. Vol. 5. Graduate Texts in Mathematics. New York: Springer, 1998.
- [Pea09] Judea Pearl. *Causality: Models, Reasoning, and Inference*. 2nd. Standard reference for causal inference and causal graphs. Cambridge: Cambridge University Press, 2009.

- [Per26] Johnny Andrey Perez Ramirez. “A Structural Criterion for Substrate-Invariant Operational Consciousness”. In: *arXiv preprint* (2026).